BCH 353

Protein Isolation: Fractionation, Purification, and Characterisation

Protein isolation is a series of processes intended to separate one or a few proteins from a complex mixture, usually cells (i.e. microbial cells and fractionated), tissues or whole organisms.
Protein isolation is essential for:

1. the analysis of the biological properties of the protein of interest,
2. to understand its structure,
3. and study its interactions.

The isolation process may separate the protein and non-protein parts of the mixture, and finally separate the desired protein from all other proteins. Separation of one protein from all others is typically the most laborious aspect of protein isolation. Separation steps usually exploit differences in protein size, physico-chemical properties, binding affinity, and biological activity. The pure result may be termed protein isolate.

The source of a protein is generally tissue or microbial cells. Prior to protein extraction, it is important to know the location of the protein within the cell, and whether the protein is present in free solution or bound to a membrane. The first step in any protein purification procedure is to break open these cells, releasing their proteins into a solution called a crude extract. Soluble cytoplasmic proteins are simplest to extract, for any disruption of the plasma membrane will enable the release of proteins to its surrounding environment, Soluble proteins present in the organelles of eukaryotic cell are also easy to liberate but the disruption of intracellular membranes often require harsher conditions than disruption of plasma membrane. To minimize the extraction of unwanted protein, subcellular fractions are prepared to isolate specific organelles. Once the extract or organelle preparation is ready, various methods are available for purifying one or more of the proteins it contains. Commonly, the extract is subjected to treatments that separate the proteins into different fractions based on a property such as size or charge, a process referred to as fractionation.

1.0 Extraction of protein- *Important considerations*

1.1 Temperature

Generally both extraction and purification processes should be done under a cold condition, mostly at 0-4 ^oC, except for some proteins. An ice box or another cooling system is normally needed for sample cooling. The extraction procedure varies according to the type of sample and physicochemical properties of the protein.

1.2 Buffer

Extraction buffer or solvent is also important. Buffer with suitable concentration, ionic strength, and pH is used for extraction of water-soluble protein.

1.3 Protease inhibitors

Protease inhibitors are sometimes necessary to prevent destruction of the extracted proteins by proteases.

1.4 Cell disruptions

The first step is to disrupt cells or tissues. The more gentle procedure is used the more intact protein is obtained. Most of animal cells and tissues are soft and easy to break, so gentle methods can be applied. For bacteria, fungi and most plant cells which have tough cell envelope, more vigorous procedures are required. From the fact that the protein extract obtained from subcellular component has contaminants less than that obtained from whole cell and will be easier to be purified. So separation of subcellular components may be done before the protein extraction. Disruption of cells should be adjusted to get intact sub-cellular components. The required cell component is obtained by centrifugation under appropriate condition.

2.0 Methods of cell disruptions

There are many methods, which have been established to disrupt cells or tissues in the macro, and micro scale and these methods can be categorized mainly as mechanical and non-mechanical techniques (Figure 1). They include:

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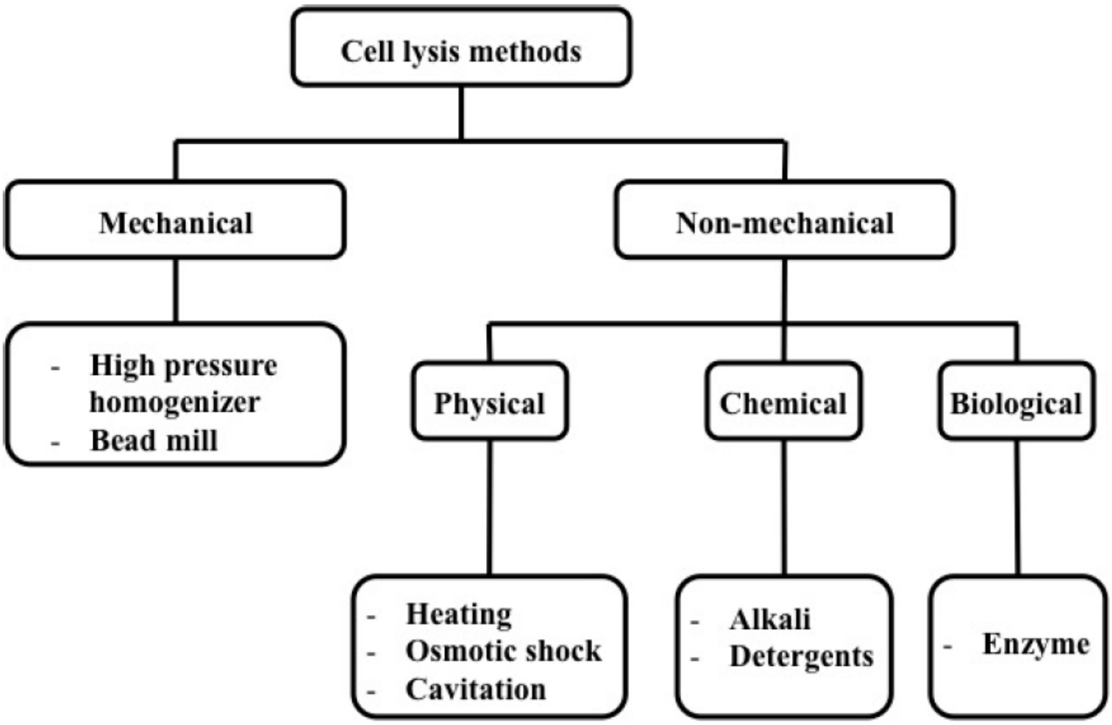

Figure 1: Methods of cell disruptions

2.1 Mechanical Lysis

In mechanical lysis, cell membrane is physically broken down by using shear force. This method is the most popular and is available commercially because of a combination of high throughput and higher lysing efficiency. Different types of mechanical lysis techniques are discussed below:

2.1.1 Homogenizer

High Pressure Homogenizer (HPH) is one of the most widely used equipment for large scale microbial disruption. In this method, cells in media are forced through an orifice valve using high pressure. Disruption of the membrane occurs due to high shear force at the orifice when the cell is subjected to compression while entering the orifice and expansion upon discharge. Hand homogenizers or dounce homogenizers are used at laboratory scale for soft tissue homogenization.

2.1.2 Bead mill

Bead mill, also known as bead beating method, is a widely used laboratory scale mechanical cell lysis method. The cells are disrupted by agitating tiny beads made of glass, steel or ceramic which are mixed along with the cell suspension at high speeds. The beads collide with the cells breaking open the cell membrane and releasing the intracellular components by shear force. This process is influenced by many parameters such as bead diameter and density, cell concentration and speed of agitator.

2.1.3 French press

French press uses shear force to homogenize the tissue. An example of this would be taking a cell suspension and forcing it through a narrow gauge syringe using a syringe barrel and plunger. This works well with bacteria, yeast, fungi, algae and mammalian cell culture.

2.1.4 Grinding

Grinding involves using a mortar and pestle to homogenize the tissue sample. The most common method is to freeze and grind the sample using liquid nitrogen. It can also be done using a non-binding abrasive. This method is effective for virtually all tissue types including seeds.

2.2 Non-Mechanical Lysis

Non-mechanical lysis can be categorized into three main groups, namely physical, chemical and biological, where each group is further classified based on the specific techniques and methods used for lysis.

2.2.1 Physical Disruption

Physical disruption is a non-contact method which utilize external force to rupture the cell membrane. The different forces include heat, pressure and sound energy. They can be classified as thermal lysis, cavitation and osmotic shock.

2.2.1.1 Thermal Lysis (Freezing and Thawing)

Cell lysis can be conducted by repeated freezing and thawing cycles. When cells are frozen the water inside the cells expands as it freezes causing the cells to burst open which helps in breaking down the cell membrane. This method is time consuming and cannot be used for extracting cellular components sensitive to temperature.

2.2.1.2 Cavitation (Sonication)

Cavitation is a technique which is used for the formation and subsequent rupture of cavities or bubbles. These cavities can be formed by reducing the local pressure which can be done by increasing the velocity, ultrasonic vibration, etc. Subsequently, reduction of pressure causes the collapse of the cavity or bubble. During the collapse of a bubble, a large amount of mechanical energy is released in the form of a shockwave that propagates through the media. Since this shock wave has high energy, it has been used to disintegrate the cell membrane. Ultrasonic and hydrodynamic methods have been used for generating cavitation used to disrupt cells.

2.2.1.3 Osmotic Shock

When the concentration of salt surrounding a cell is suddenly changed such that there is a concentration difference between the inside and outside of the cell, the cell membrane becomes permeable to water due to osmosis. If the concentration of salt is lower in the surrounding solution, water enters the cell and the cell swells up and subsequently bursts. This technique is suitable for mammalian cell due to the fragile structure of membrane; however, periplasmic proteins may be released in the case of gram-negative bacteria.

2.2.2 **Chemical Cell Disruption**

Chemical lysis methods use lysis buffers to disrupt the cell membrane. Lysis buffers break the cell membrane by changing the pH. Detergents can also be added to cell lysis buffers to solubilize the membrane proteins and to rupture the cell membrane to release its contents. Chemical lysis can be classified as alkaline lysis and detergent lysis.

2.2.2.1 Alkaline Lysis

In alkaline lysis, OH^- ions are the main component used for lysing cell membrane. The lysis buffer consists of sodium hydroxide and sodium dodecyl sulphate (SDS). The OH^- ion reacts with the cell membrane and breaks the fatty acid-glycerol ester bonds and subsequently makes the cell membrane permeable and the SDS solubilizes the proteins and the membrane. The pH range of 11.5–12.5 is preferable for cell lysis. Although this method is suitable for all kinds of cells, this process is very slow and takes about 6 to 12 h.

2.2.2.2 Detergent Lysis

Detergents also called surfactants have an ability to disrupt the hydrophobic-hydrophilic interactions. Since the cell membrane is a bi-lipid layer made of both hydrophobic and hydrophilic molecules, detergents can be used to disintegrate them. Detergents are capable of disrupting the lipid–lipid, lipid–protein and protein-protein interactions. Based on their charge carrying capacity, they can be divided into cationic, anionic and non-ionic detergents. Detergents are most widely used for lysing mammalian cells. For lysing bacterial cells, first the cell wall has to be broken down in order to access the cell membrane. Detergents are often used along with lysozymes for lysing bacteria (e.g., yeast). Out of the three types of detergents, non-ionic detergents are mostly preferred as they cause the least amount of damage to proteins and enzymes. 3-[(3-cholamidopropyl)dimethylammonio]-1-propanesulfonate (CHAPS) and 3-[(3-cholamidopropyl)dimethylammonio]-2-hydroxy-1-propanesulfonate (CHAPSO),a zwitterionic detergent, is one of the most popular non-ionic detergents. Other non-ionic detergents include Triton-X and Tween series.

2.2.3 Biological Cell Disruption

2.2.3.1 Enzymatic Cell Lysis

Enzymatic lysis is a biological cell lysis method in which enzymes such as lysozyme, lysostaphin, zymolase, cellulose, protease or glycanase are used. Most of these enzymes are available commercially and can be used for large scale lysis. One advantage of enzymatic lysis is its specificity. For example, lysozymes are used for bacterial cell lysis whereas chitinase can be used for yeast cell lysis and pectinases are used for plant cell lysis. Lysozyme reacts with peptidoglycan layer and breaks the glycosidic bond. For that reason, gram-positive bacteria can be directly exposed to lysozyme, however, outer membrane of the gram-negative bacteria needs to be removed before exposing the peptidoglycan layer to the enzyme. Lysozyme treatment is generally conducted at pH 6–7 and at 35 ^{o}C. For gram-negative bacteria, lysozyme is used in combination with detergents to break the
cell wall and membrane.

2.3 Extraction of water-soluble protein from animal cells and tissues

The proteins which are components of fragile unicellular tissues such as animal blood cells can be extracted by using hypotonic buffer solution. If the sample contains different types of cells, separation of the cell types before the extraction will make the purification easier. This osmosis-based method is also used for animal cells grown in culture media. In some case, a mild surfactant is included in the extraction buffer. Sonication or freeze/thaw cycle or a mild mechanical agitation may be used to help disruption of the cells and dissociation of the cellular components. Cell disruption by nitrogen decompression is another method for animal cells and some plant cells. In this method, large quantities of nitrogen are dissolved in the cells under high pressure in a vessel. When the pressure is suddenly released, the dissolved nitrogen becomes bubbles. The expanding bubbles cause rupture of the cell. Intact organelles are also obtained by this method.

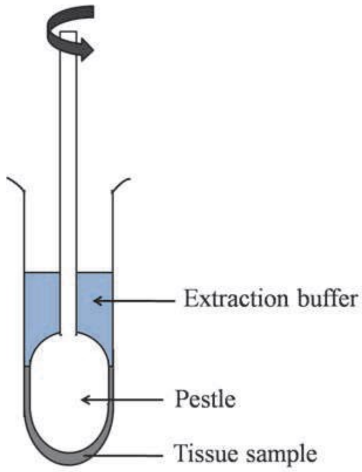

Figure 2. Homogenization of tissue by a hand homogenizer.

For multicellular soft tissue, a hand homogenizer (Figure 2) or an electrical one with optimal size is a good choice. The tiny pestle using with a micro-centrifuge tube is also commercially available for the sample with a small volume. Equipment with stronger breaking power such as a blade blender are needed for extraction of proteins from tougher tissue such as animal muscle. This treatment ruptures the plasma membrane but leaves most of the organelles intact. The homogenate is then centrifuged; organelles such as nuclei, mitochondria, and lysosomes differ in size and therefore sediment at different rates. They also differ in specific gravity, and they “float” at different levels in a density gradient.

Differential centrifugation results in a rough fractionation of the cytoplasmic contents, which may be further purified by isopycnic (“same density”) centrifugation (Figure 3a). In isopycnic centrifugation, a centrifuge tube is filled with a solution, the density of which increases from top to bottom; a solute such as sucrose is dissolved at different concentrations to produce the density gradient. When a mixture of organelles is layered on top of the density gradient and the tube is centrifuged at high speed, individual organelles sediment until their buoyant density exactly matches that in the gradient. Each layer can be collected separately (Figure 3b).

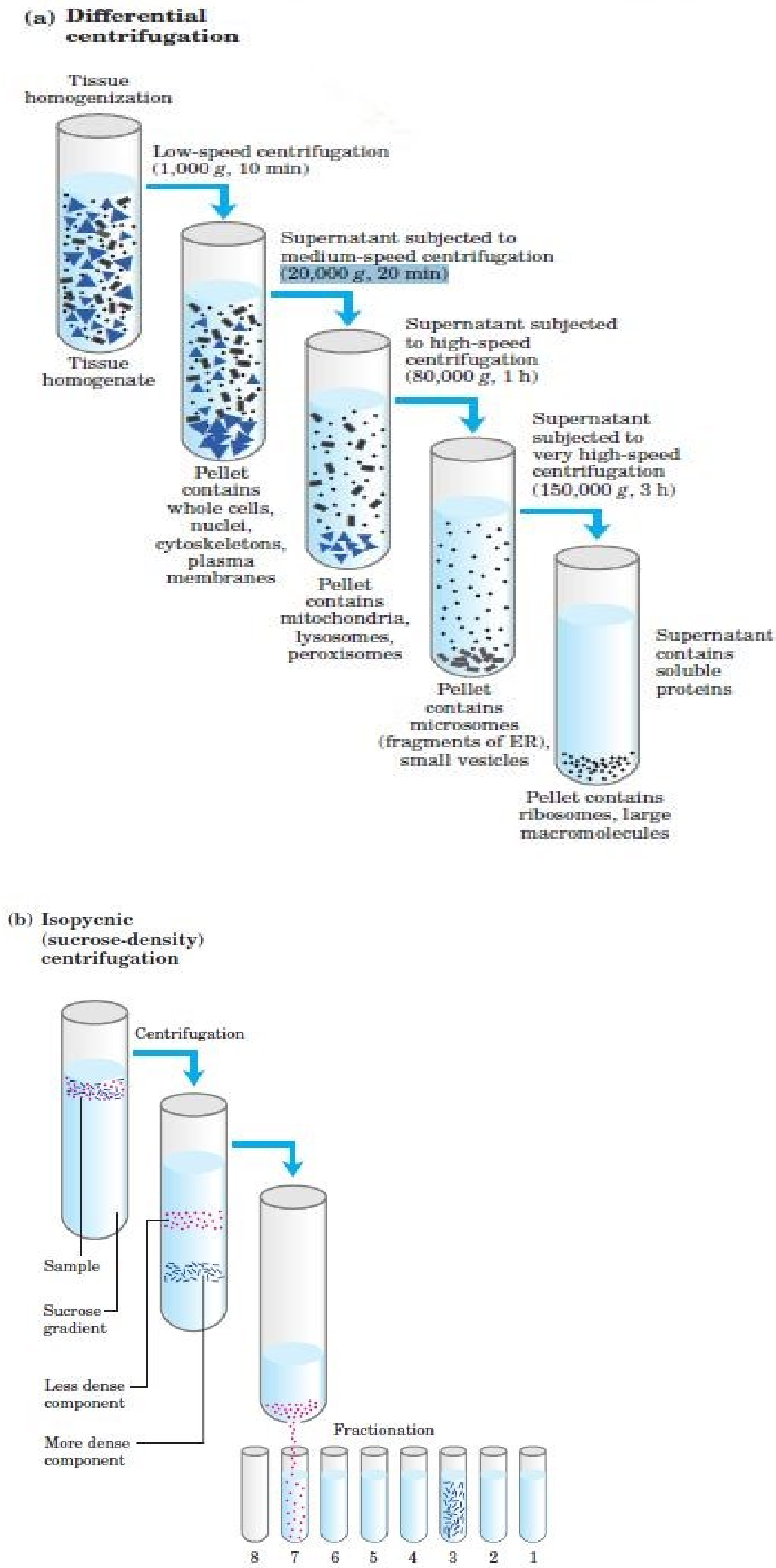

Figure 3: Differential centrifugation (a) and Isopycnic centrifugation (b)

2.4 Extraction of water-soluble protein from unicellular organisms

This group of organisms includes bacteria, yeast, fungi and some algae. Their cell envelopes are tougher than those of animal cells. The stronger methods are needed for disruption of the cells. A simple method is to shake the suspension of cells, as well as spores, with small glass beads or the other kind of beads. If the collision with the beads is done in a blender, the method is called a bead mill. Sonication is another way to break the cells. The solution used to suspend the cells can be only an appropriate buffer, or with the addition of some enzyme and/or mild detergent. The examples of the enzymes are lysozyme, cellulase and chitinase. Nonionic and zwitterionic detergents, especially Triton X-100 and CHAPS (3-[(3-cholamidopropyl) dimethylammonio]-1-propanesulfonate), are commonly used since they are mild and have little effects on protein denaturation. To disrupt the unicellular organisms with very strong envelope, a high-pressure press apparatus may be needed such as French press, Microfluidizer processors. In these systems, the cell suspension in a suitable chamber is pressurized (up to 30,000 psi) by using a hydraulic pump. Shearing force is generated when the pressurized suspension is squeezed pass a very narrow outlet into the atmospheric pressure.

2.5 Extraction of water-soluble protein from plant tissue

Grinding with or without sands, in the presence of an appropriate buffer, by using a pestle and mortar actually works well with plant sample. If the plant tissue is too strong to be ground, rapid freeze with liquid nitrogen will make the plant more fragile. The frozen plant tissue is ground before shaking in the buffer. Some plant cells can be disrupted by the mean of nitrogen decompression.

2.6 Extraction of lipid-soluble protein

Most of lipid-soluble proteins are membrane proteins. These proteins may be called proteolipids which are extracted from the samples by using organic solvents such as a mixture of chloroform and methanol. An aqueous solution containing mild detergents such as Triton X-100 and CHAPS is an alternative way to dissolve the proteins from cells or organelles. Strong detergents can be used but they may cause permanent denaturation of the proteins.

2.7 Extraction of aggregated protein

High expression of recombinant proteins in bacterial host cell often form insoluble aggregates which is known as inclusion bodies. This form of proteins is very difficult to be solubilized. To extract any protein from bacterial inclusion bodies, strong denaturants such as 6M guanidine-HCl or 6 to 8M urea are included in the solubilizing solution. This solution efficiently extracts the aggregated protein. However, the extracted protein is also denatured and sometimes cannot be renatured. The other procedure is to use a mixture of mild detergents such as a combination of Triton X-100, CHAPS and sarkosyl. This method is less efficiency but more native forms of the protein can be recovered.

Summary of protein extraction methods from different sources

Animal cells

- Hypotonic buffer solution (0.45 % of normal saline)
- Sonication
- Freeze/thawing
- Mild mechanical agitation
- Nitrogen decompression

Animal tissues

- Homogenization using a homogenizer (soft tissues)
- Blending using Waring Blender (for tough tissues)

Microbial Cells

- Bead mill
- Sonication
- Lytic enzymes such as lysozymes, chitinase and cellulose
- Mild detergent – Triton X, CHAPS
- French press- disrupt organisms with strong envelope

Plant cell

- Grinding with or without sand – in the presence of appropriate buffer solution
- Rapid freeze with liquid Nitrogen
- Enzymatic lysis using pectinase

Lipid soluble protein

- Mixture of organic solvents such as methanol and chloroform
- Aquous solution containing detergents

Aggregated proteins

- Strong denaturants such as 6M guanidine-HCl or 6-8M urea are included in the solubilizing buffer
- Detergents such as Triton X

PROTEIN PURIFICATION

Purpose of protein purification

There are two major purposes for protein purification. It could be either for *preparative* or *analytical purpose*.

Preparative purifications- aim to produce a relatively large quantity of purified proteins for subsequent use. Examples include the preparation of commercial products such as enzymes (e.g. lactase), nutritional proteins (e.g. soy protein isolate), and certain biopharmaceuticals (e.g. insulin). Several preparative purifications steps are often deployed to remove bi-products, such as host cell proteins, which poses as a potential threat to the patient's health.

Analytical purification produces a relatively small amount of a protein for a variety of research or analytical purposes, including identification, quantification, and studies of the protein's structure, post-translational modifications and function. Pepsin and urease were the first proteins purified to the point that they could be crystallized.

Purification Methods

After the preparation of crude extract, various methods for the purification of one or more proteins can be applied. Usually, the extract is subjected to preparations that separate the proteins into different fractions based on a particular property such as solubility, size, charge, specific binding affinity, hydrophobicity and isoelectric point. That process is known as fractionation. All of these properties can be exploited to separate them from one another so that they can be studied individually. The most essential methods for the protein purification and the protein properties that they are based on are listed on Table 1.

Table 1. Classification of purification techniques according to protein properties

Technique	Protein Property
Precipitation with Ammonium sulfate (Salting out)	Solubility
Ion-exchange Chromatography (IEC)	Charge
Gel filtration Chromatography (GF)	Size
Affinity Chromatography (AC)	Ligand Specificity
Hydrophobic Chromatography (HIC)	Hydrophobicity
Reversed Phase Chromatography (RPC)	Hydrophobicity
Chromatofocusing	Isoelectric point

Precipitation with Ammonium Sulfate

The solubility of proteins is generally lowered at high salt concentrations, an effect called "salting out." The addition of a salt in a specific quantity can effectively precipitate some proteins, while others remain in solution. Hence, salting out can be used to fractionate proteins in the first steps of purification process. Ammonium sulphate ($(NH_4)_2SO_4$) is a common salt used for this process because of its high solubility in water, its low cost. It does not affect the solution's temperature and does not contribute to the denaturation of protein. Salting out is also useful for concentrating dilute solutions of proteins, including active fractions obtained from other purification steps. Dialysis can be used to remove the salt if necessary. It is a wide used method because of its simple equipment and low cost.

Ion-Exchange Chromatography

Protein purification can be achieved by ion-exchange chromatography (IEC) on the basis of their net charge giving a very high resolution separation with high sample loading capacity. The principle of separation is based on the reversible interaction between a charged protein and an oppositely charged chromatographic medium. Proteins' binding is achieved while they are loaded onto a column. Then conditions are altered so that bound materials are eluted in a different way by performing changes in salt concentration or in pH. Changes are made in one step or with a continuous gradient. In most cases, proteins are eluted with salt (NaCl), using a gradient elution increasing the ionic strength of the eluent. The selectivity of this method is affected by column, mobile-phase buffer, counter-ion salt type, gradient steepness, other mobile phase additives (such as surfactants) and temperature. The net surface charge of proteins varies according to the surrounding pH. When pH is above its isoelectric point (pI) a protein will bind to an anion exchanger, when below its pI a protein will bind to a cation exchanger. Cation exchangers contain acidic groups that are referred to as weak (e.g.,$-COO^-$) or strong (e.g., $-SO_3^-$), whereas anion exchangers contain basic groups that may be weak (e.g., $-NH_2$) or strong (e.g., $-NR_3^+$).

Specifically, proteins that have a low density of net positive charge will tend to emerge first, followed by those having a higher charge density (Figure 4). As a guideline, anion exchange separations are often carried out at 1 to 1.5 pH units above a protein's pI, and cation-exchange separations at 1 to 1.5 pH units below the pI. Positively charged proteins (cationic proteins) can be separated on negatively charged carboxymethyl-cellulose (CM-cellulose) columns. Conversely, negatively charged proteins (anionic proteins) can be separated by chromatography on positively charged diethylaminoethyl-cellulose (DEAE-cellulose) columns. Beyond cellulose, agarose, dextran, silica and synthetic polymers can be used for packing a column. Furthermore, ion-exchange chromatography can be used to bind impurities if required. It can be repeated at different pH values to separate several proteins, which have distinctly different charge properties. The reasons for the success of ion exchange are its widespread applicability, its high resolving power, its high capacity, and the simplicity and controllability of the method.

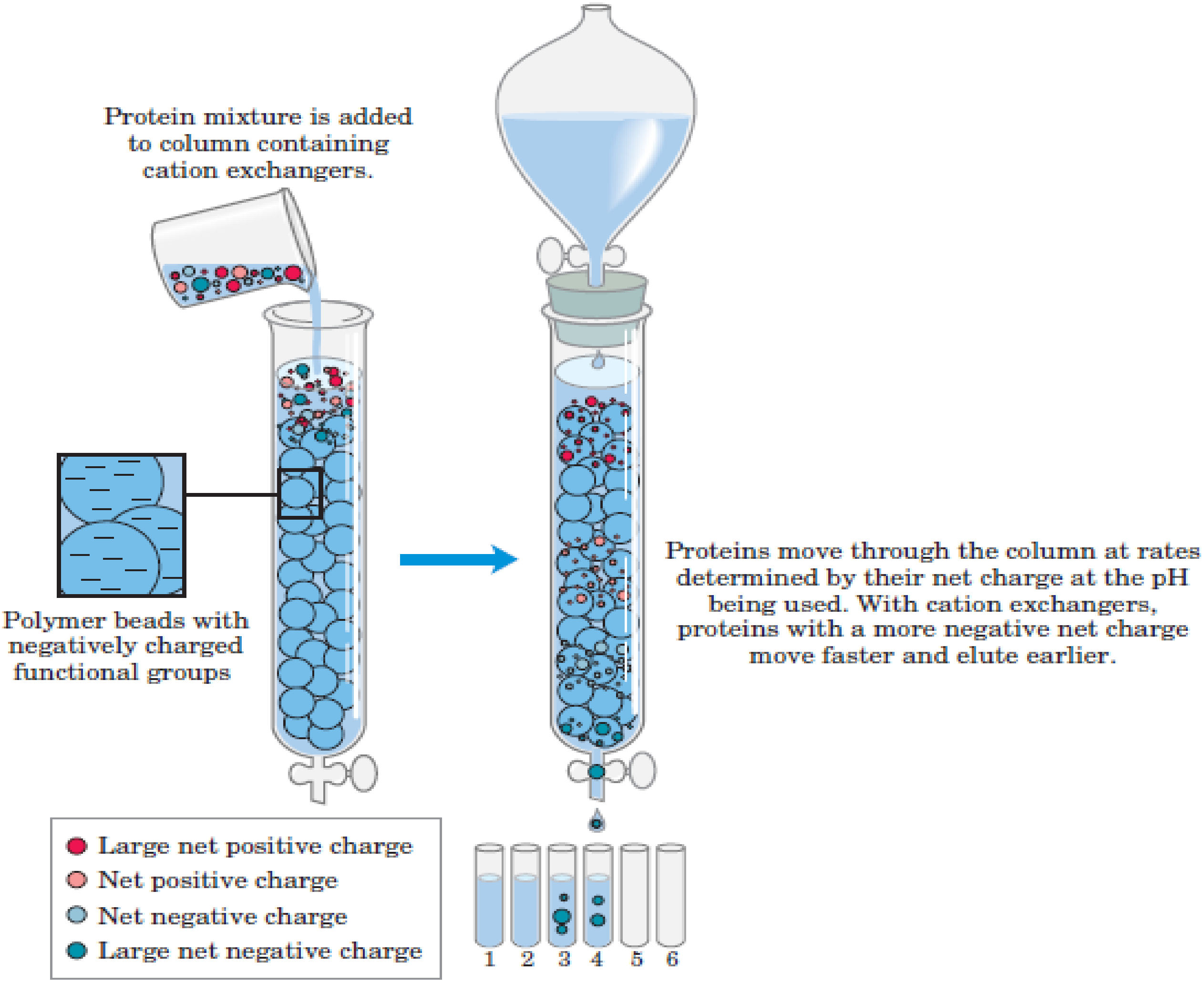

Figure 4. Ion exchange chromatography as a protein purification method

Size-Exclusion Chromatography

Separation of proteins in Size-Exclusion Chromatography (SEC) is based on the molecule size rather than any interaction phenomena. In this method, large proteins emerge from the column sooner than smaller ones. The column is packed with a cross-linked polymer. Especially, it consists of beads with engineered pores of a particular size. Large proteins cannot enter the pores and so they take a more direct path through the column, around the beads. However, if proteins are small, they enter the pores and pass the column more slowly as a result. Hence, large proteins appear in the earlier fractions during the elution. Figure 5 describes the above process of purification. Before the application of this method, it is important to take into consideration that the protein sample must be dissolved well in the mobile phase and it must not interact with the polymeric stationary phase. Furthermore, in order to achieve a simple and fast separation the mobile phase must be suitable for the stationary phase without disrupting it. Size- Exclusion Chromatography is also called Gel Filtration (GF) when the mobile phase is aqueous.

Gel Filtration is a powerful method for separating proteins with extremes in size. Because the supports of this column are non-interactive, they have no trace-enrichment potential and cannot be used for the purpose of concentrating proteins. Furthermore, it is not usually used as a first purification step, but it is useful for removing impurities, oligomers and aggregates of the target protein

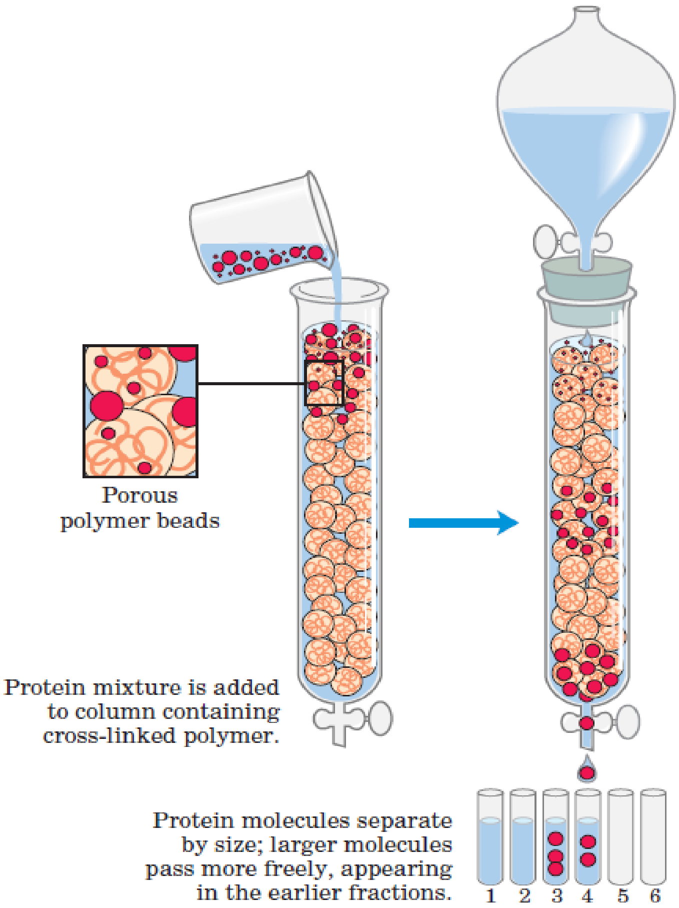

Figure 5. Size exclusion chromatography as a protein purification method.

Affinity Chromatography

Affinity Chromatography (AF) is one of the most powerful and widely applicable chromatographic methods for protein purification. Generally, it can be used for purification of a specific molecule from complex mixtures. It is based on the use of an immobilized natural ligand creating a stationary phase, which specifically interacts with the desired protein. The protein-target is in a mobile phase as part of a complex mixture. The mobile phase with desired protein is passed into the column, and the specific interaction holds back the protein of interest, while others are washed through the column, as it is represented in Figure 6. After this, the bound protein is eluted by a solution. The effectiveness of affinity purification depends on various factors, including the available amount of the ligand, the accessibility of the ligand on the resin, the strength of the interaction, and the integrity of protein to be immobilized. The optimization of the whole process is required in order to maximize the interaction between a protein-target and an immobilized ligand during the binding and wash step. Then, the interaction becomes less weak allowing the release of the protein.

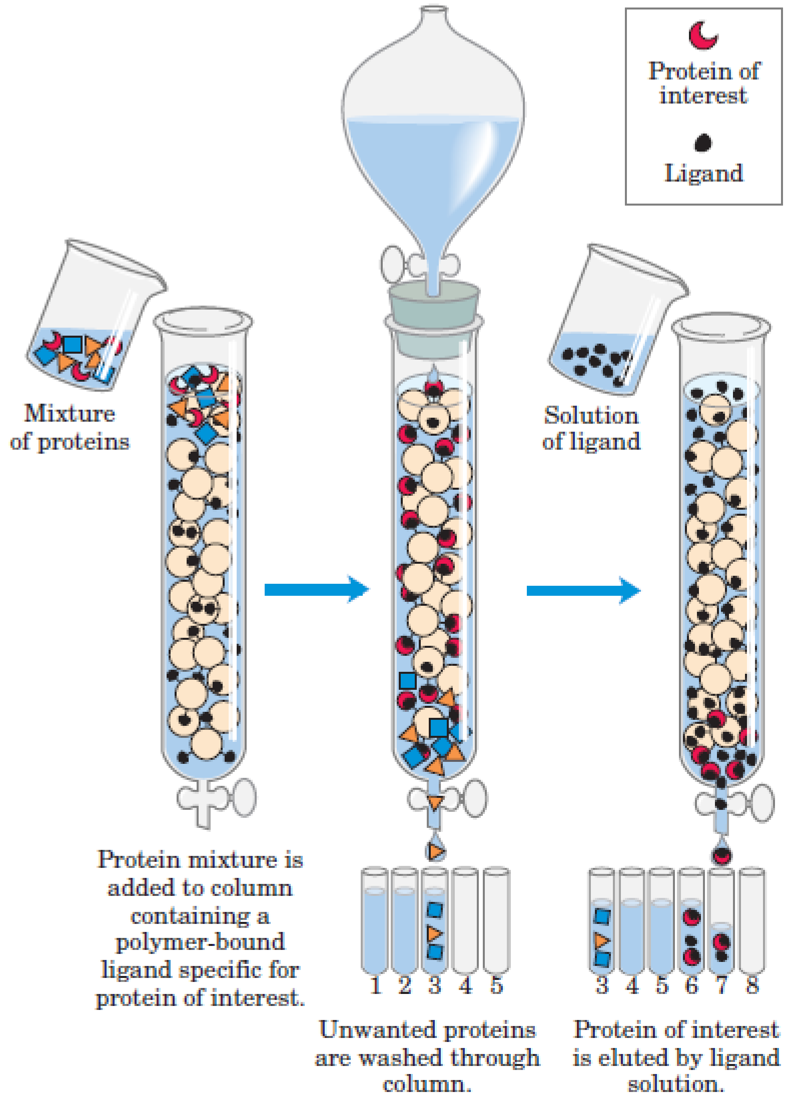

Figure 7. Affinity Chromatography as a protein purification method

Hydrophobic Interaction Chromatography

Hydrophobic Interaction Chromatography (HIC) was first described by Tiselius in 1940s. Since the 1960s, it has been for separation of proteins with carbohydrate-based packings, and more recently with high-performance microparticulate supports. The principle of this method is based on the interaction between proteins and hydrophobic surface of a chromatography medium in the presence of high concentration of salt. Hydrophobic interaction, between the protein and the hydrophobic ligand is driven primarily by an increase in the overall entropy (compared with the condition when no interaction is occurring between the protein and the adsorbent). During separation, the concentration of salts decreases in a reverse gradient elution. As a result, the strength of hydrophobic interaction between the protein and the ligand decreases, and the protein will be desorbed and eluted from the column.

Hydrophobic interaction chromatography is a gentle technique, with proteins eluting in their native conformation without losing their biological activity. Also, it is widely used for the preparative isolation of proteins in laboratory scale-up to process-scale applications. This method is used to remove various impurities that may be present in the solution, including undesirable product-related impurities. In particular, HIC is often used to remove product aggregate species, which possess different hydrophobic properties than the target monomer species. Adsorbents in HIC are silica and polymer based and they consist of porous beads with diameters between 5 to 200 μm, a satisfying area for ligand attachment and protein binding. The strength of hydrophobic binding of the ligand increases with the length of the organic chain. Moreover, the hydrophobic interaction strength can be affected by ligand density on the support. The strength becomes higher with higher ligand densities.

Reversed Phase Chromatography

Reversed Phase Chromatography (RPC) is one of the most important and powerful techniques for protein purification. Several features of this technique are responsible for its great appeal among the purification techniques. Especially, RPC columns are packed with small particles redounding to increased resolution, versatility, stability, reproducibility and high peak capacities for the separation of complex mixtures. Furthermore, it’s compatibility with mass spectrometry makes it part and parcel of protein studies.

As in Hydrophobic Interaction Chromatography, the separation in RPC is based on the hydrophobicity of the protein molecule. RPC columns are mainly packed with silica gel or polymers. Especially, for separation of proteins large pore silicas are required (pore diameter ≥30) for high efficiency. Also, monolithic columns are an alternative choice avoiding bulk materials. These columns are made from silica or polymer and have a rigid cylinder shape. The most popular stationary phases are based on straight-chain alkyl groups (C4, C8, C18).

In RPC the mobile phase consists of an aqueous buffer, often a dilute acid such as phosphoric, trifluoroacetic, formic or acetic acid, and an organic solvent, such as methanol, acetonitrile or isopropanol. It is important to maintain pH of aqueous buffer in a stable range from 2 to 3.5 avoiding the denaturation of proteins. The presence of acid renders the proteins positively charged and minimizes unpleasant interaction with the stationary phase.

Electrophoresis

Electrophoresis is another important technique for protein purification introduced in the beginning of 20th century. The whole process is based on the migration of charged proteins in an electric field at a constant pH and current. This technique provides the advantage of visualizing the proteins during the separation. This allows the rapid estimation of the quantity of proteins in a mixture and the yield of purity of each protein separation. Furthermore, the velocity of migration of proteins reflects their isoelectric point and approximate molecular weight The velocity of migration (v) of a protein in an electric field depends on the electric field strength (E), the net charge on the protein (z) and the frictional coefficient (f) according to the Equation 1.

$$v = \frac{Ez}{f} \qquad (1)$$

This force is opposed by viscous forces in the medium, proportional to viscosity η, particle radius r (Stokes radius), and the velocity (v) as it is shown on the Equation 2.

$$f = 6\pi\eta r \qquad (2)$$

There are two types of electrophoresis, one and two dimensional depending on the scale of separation. One dimensional electrophoresis is applied for most routine protein purification. Separations by electrophoresis are carried out in thin and vertical gels made up of the cross-linked polymer polyacrylamide because they are chemically inert and are readily formed. The direction of flow is from top to bottom. Molecules that are small compared with the pores in the gel readily move through the gel, whereas molecules much larger than the pores stay at the top. Intermediate-size molecules move through the gel with various degrees of facility.

The most common applied electrophoretic method for the estimation of purity and molecular weight is based on the detergent sodium dodecyl sulfate (SDS) (Figure 8). SDS is an anionic detergent, meaning that when dissolved its molecules have a net negative charge within a wide pH range. A polypeptide chain binds amounts of SDS in proportion to its relative molecular mass. SDS-electrophoresis separates proteins on the basis of their molecular weight. Thus, the progress of the purification process can be monitored as the number of protein bands visible on the gel decreases after each new fractionation step. If the position of a protein with known molecular weight in the gel is known, it is possible to measure the molecular weight of an unidentified protein by monitoring its position. If the protein consists of two or more different subunits, the subunits will generally be separated by the SDS detergent and a separate band will appear for each.

Isoelectric Focusing

Isoelectric focusing is an electrophoretic method where proteins are separated on the basis of their isoelectric point (pI). The pI is the pH at which a protein has no charge and, thus, does not migrate further in an electric field. By the presence of a pH gradient, a mixture of low molecular weight organic acids and bases is applied in an electric field generated across the gel. Each protein migrates throughout the gel until it meets its pI (Table 2).

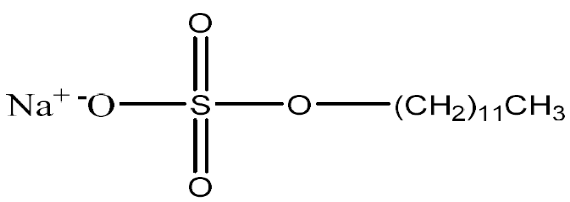

Figure 8. Structure of sodium dodecyl sulfate (SDS).

Table 2. The isoelectric points of some proteins

Protein	pI
Pepsin	<1.0
Egg albumin	4.6
Serum albumin	4.9
Urease	5.0
β-Lactoglobulin	5.2
Phycoerythrin	5.74
Hemoglobin	6.8
Myoglobin	7.0
Chymotrypsinogen	9.5
Cytochrome c	10.7
Lysozyme	11.0

Two-Dimensional Electrophoresis

There are some complex mixtures of proteins that cannot be separated by one dimensional gel electrophoresis. In these cases, two-dimensional electrophoresis is applied, which combines isoelectric focusing and SDS electrophoresis sequentially in a process achieving the separation of more than one hundred proteins in a polyacrylamide gel. Two-dimensional polyacrylamide gel electrophoresis (2D-PAGE) is usually performed with isoelectric focusing in the first dimension and SDS-PAGE in the second dimension. Especially, this method separates proteins of identical molecular weight in the vertical direction that differ in pI, or proteins with similar pI values but different molecular weights in the horizontal direction.

ANALYSIS AND CHARACTERIZATION OF SEPARATED PROTEINS

In a protein purification process it is crucial to determine the purity and the concentration of each purification step. Thus, researchers applied some specific techniques in order to accomplish the above requirements. Simultaneously, these techniques provide information about the presence of impurities and assess the protein structure.

Spectroscopy

To quantify the amount and concentration of purified protein, the simplest and most common method is the use of a spectrophotometer, since proteins absorb light at a specific wavelength. Most proteins have an absorption maximum at about 280 nm, which is due to the aromatic amino acids. The type of amino acids that are present in a protein affect the absorption. Especially, proteins that have similar molecular weight but different amount of tyrosine and tryptophan will have different absorbance values. Furthermore, there are some other parameters that may affect the absorbance of a specific protein, such as temperature, ionic strength, pH and the presence of detergents. The integrity of amino acids depends on the above parameters, so the ability of aromatic residues to absorb at 280 nm will be affected, changing the value of the protein's extinction coefficient. This method of detection can be used independently in the stand-alone mode using a manual syringe or pump for sample injection or it can be used on-line with an HPLC system.

After a purification step it is recommended to assess the purity by a spectrophotometer obtaining quick results without destructing the protein sample. Apart from UV-visible spectroscopy, fluorescence spectroscopy is used for the quantitation and characterization of purified proteins. Especially, the fluorescence emission signal is measured at 280 nm and 340 nm, corresponding respectively to light scattering and intrinsic protein fluorescence, after excitation at 280 nm. The ratio of the intensities at 280 nm and 340 nm (I280/I340) is called aggregation index (AI), because it is related to the degree of aggregation of the sample, but not to the concentration. This index should have values close to zero for aggregate-free protein preparations and can obtain values higher than 1 unit, when significant aggregation takes place

Gel Filtration for the Determination of the Molecular mass

As it was mentioned before, gel filtration chromatography is a preparative, non-destructive analytical technique that permits the separation of proteins by their size in a purification process. Besides protein purification, this technique can be used for determining the molecular weight of proteins. The molecular mass of a given protein may be determined by comparing its ratio of Ve/Vo (Ve: elution volume, Vo: void volume) with those of protein standards with known molecular mass, which are commercially available. The column void volume corresponds to the elution volume of a very large molecule. Plotting the logarithms of the known molecular masses of protein standards against their respective ratios of elution volume to column void volume, Ve/Vo values, produces a linear calibration curve.

Mass Spectrometry

Nowadays, mass spectrometer detectors has become the most popular detector systems for bioanalytical methods, such as purification processes. MS detectors are used in two conformations, the mass selective detector (MSD) and the multiple reaction monitoring (MRM). The first type of detection is based on the measurement of the protonated molecular ion (M + H) for each analyte. Whereas, the MRM process is based on the isolation of the primary ionic species (parent ions), the fragmentation of them into additional ions (product ions) and the monitoring of them. When mass spectrometry is combined with HPLC systems, where proteins have been separated, the systems are called LC-MS or LC-MS/MS, respectively. The MS detector interface disperses the liquid sample of proteins into the gas phase, whereas the pressure must be extremely lower than the atmospheric (760 Torr), in a range of 10-5 to 10-6 Torr. The most popular ionization methods are electrospray ionization (ESI), atmospheric pressure chemical ionization (APCI) and matrix-assisted laser desorption-ionization (MALDI).

One of the most widely used MS detector is the time-of-flight (TOF). In this mass analyzer, ions are derived from the ion source and are accelerated with a specific energy and directed through a high vacuum flight tube to the detector. The small ions are faster than heavy, so they reach detector first, since they demand less energy. The time of flight of ions is related to the mass (m/z) of the ion.

Now, MALDI-TOF-MS is routinely used in many laboratories for the rapid and sensitive identification of proteins by peptide mass fingerprinting (PMF). Firstly, proteins are separated by one- or two-dimensional gel electrophoresis or liquid chromatography. Then, proteins are de-stained and cleaved with sequence-specific endoproteases, such as an in-gel trypsin digestion. Proteins can be identified at femtomole levels by MALDITOF- MS and database searching of the PMF data.

Circular Dichroism

Circular Dichroism is a supplementary technique that is used for the confirmation of the integrity of secondary structure and folding proteins that have been obtained after purification process and characterization. It is measured by a CD spectropolarimeter, which is able to measure accurately in the far UV (180-260 nm) and the near UV (250- 310 nm). Circular dichroism is defined as the unequal absorption of left-handed and right-handed circularly polarized light from optically active molecules. The degree of the difference in absorbance depends on the wavelength of light. A circular dichroism spectra comes of serial measurements of this difference in absorbance in a specific range of the electromagnetic spectrum. The analysis of CD spectra can yield valuable information related to the secondary structure of biological macromolecules, such as proteins and peptides.

X-Ray Crystallography

Protein X-ray crystallography is a technique used to obtain the three-dimensional structure of a particular protein by x-ray diffraction of its crystallized form. This three dimensional structure is crucial to determining a protein's functionality. The specificity of the protein's active sites and binding sites is completely dependent on the protein's precise conformation. X-ray crystallography can reveal the precise three-dimensional positions of most atoms in a protein molecule because x-rays and covalent bonds have similar wavelength, and therefore currently provides the best visualization of protein structure. It was the X-ray crystallography by Rosalind E.Franklin, that made it possible for J.D. Watson and F.H.C. Crick to figure out the double-helix structure of DNA

Nowadays scientists hold many useful tools in their hands to purify and isolate proteins of interest, so their conformation, properties and specific activities can be studied. The final application of protein defines the required level of its purity. The assessment of isolation process can be achieved by a wide range of techniques, which can determine also the concentration of the purified protein. The most challenging part for scientists is to combine in the most effective way the techniques that have been described above with the minimum loss of the desired protein in the fewest steps. In general terms, isolation and purification of proteins is a field that will not stop evolving as the study of cells and organisms is limitless.